Technical Enrichment Activities (35 Marks)

These components are applicable to all project categories (Research-Based, Application-Based, and Software-Based) and are intended to strengthen the technical knowledge, project quality, documentation skills, and professional development.

Mark Distribution

Component	Nature	Marks
Professional Online Certification (Individual)	Mandatory	20
Others (Group)	Optional (Minimum 3 out of 8)	15 (5 mark each)
	Total	35

1. Professional Online Certification (20 Marks)

Each student must individually complete one approved 10–20 hour online certification relevant to the project domain.

- Certification should be from a recognized platform (e.g., NPTEL, Coursera, edX, Udemy, Google, Microsoft, AWS, IBM, Oracle, Cisco, MATLAB Academy, etc.).
- The course must be approved by the project guide and should be relevant to the project.
- Each student must submit the completion certificate, course details, and a one-page summary explaining how the course supports the project.

2. Other Activities (15 Marks)

Each project team must complete any three of the activities in the table below. Each activity carries 5 marks. All selected activities **must be related to the project** and **must receive prior approval from the respective project guide** before commencement.

#	**Deliverable**	**Description**	**Expected deliverables**
1	Submission-ready Research Paper Draft	Prepare a complete research paper in IEEE, Elsevier, Springer, ACM, or any other appropriate conference/journal format based on the project work.	Complete manuscript; Figures and tables; References; Similarity report

2	Patent Disclosure Draft	Prepare a patent disclosure document based on the novelty of the proposed project.	Prior-art search; Novelty analysis; Problem statement; Proposed invention; Complete patent disclosure document
3	Dataset Release	Release a project-specific dataset in a public repository for research reuse.	Public repository link (e.g., Kaggle, Zenodo, Figshare, GitHub); Dataset documentation (README); Metadata and annotation guidelines; License information; Version details; Organized dataset repository
4	Project Website / Documentation Portal	Develop a website or documentation portal showcasing the project.	Project objectives; Problem statement; Methodology; Team details; Project progress; Documentation; GitHub repository link; Demonstration videos
5	Open-source Technical Repository	Maintain a professional GitHub repository containing reusable project resources.	Source code; Well-structured project organization; README; Installation guide; Documentation; Regular commits with version history
6	Technical Competition / Innovation Challenge	Participate in technical competitions relevant to the project (e.g., Kaggle Competitions, Hackathons, Coding Competitions, AI/ML Challenges, Innovation Challenges, other project-related technical competitions).	Registration proof; Submission proof; Brief summary of the activity and learning outcomes
7	User Study / Field Study	Conduct a user study or field evaluation wherever applicable to the project.	User study plan; Questionnaire/feedback form; Feedback collected from participants (recommended minimum of 30 participants,

			where applicable); Analysis of the feedback; Improvements incorporated into the project
8	Working demonstration	Develop a working end-to-end prototype that demonstrates the core functionality of the project and validates key performance outcomes.	Demonstrate a working model of the project - hardware or deployed software. Also, submit a run-based demo: short video + instructions + the commands/script that regenerates the metrics and sample outputs (with model/config versions).

Instructions about the submission and evaluation of Technical Enrichment Activities (35 Marks)

1. All Technical Enrichment Activities must be completed **one week before** the last working day of the 7th semester.
2. Activities completed without prior approval from the guide may not be considered for evaluation.
3. Students are required to submit the supporting evidence and relevant documents for all completed activities through their project guide.
4. The project guide will verify the submitted evidence and upload the consolidated submissions to the designated OneDrive submission link within the stipulated deadline.
5. The supporting documents may be included, wherever applicable:
6. The panel will review the submitted evidence and supporting documents for each activity.
7. Marks will be awarded only after verification of the submitted evidence and based on the prescribed evaluation criteria.
8. Activities that are incomplete, unsupported by adequate evidence, not directly related to the project, or undertaken without prior approval from the project guide may not be considered for evaluation.
9. Students are advised to maintain all supporting documents, reports, repository links, and other evidence in an organized manner throughout the semester to facilitate smooth verification and evaluation.